Hybrid III 6 y/o Lab

BME254L - Spring 2026 - Palmeri

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2026-01-07

Table of contents

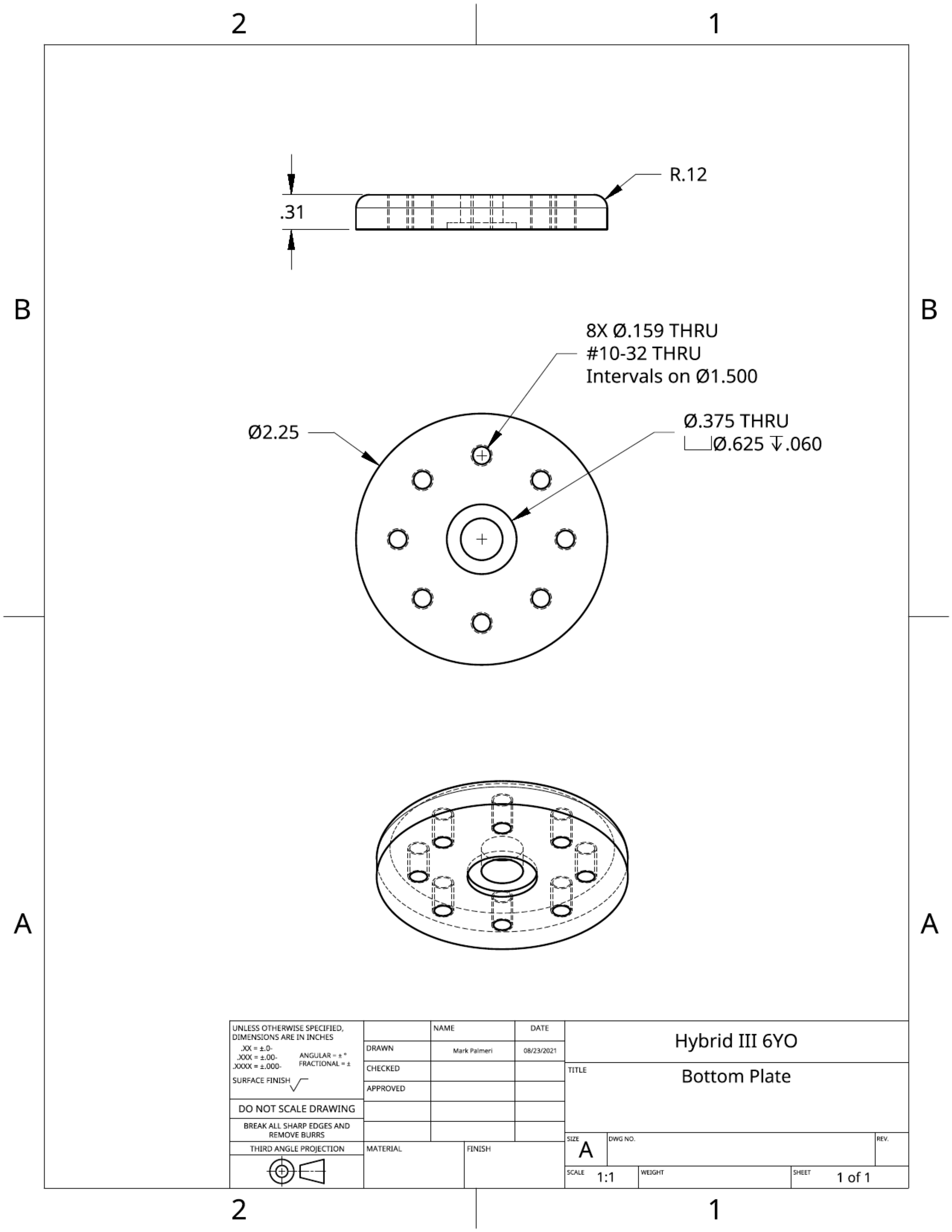
Tasks

- Create all of the parts for the Hybrid III 6yo test setup based on the attached drawings.
- You should create all of these parts and their associated drawings and assemblies as part of the same "Document" and share that document with Dr. Palmeri and the TA(s).
- Create the mechanical drawings associated with each of those parts (i.e., recreate the drawings that you are using to generate the parts) and the assembly.
- Assemble your parts as guided by the assembly diagram and as shown in the uploaded animation.
- There are online tutorials on Assemblies if you need some refreshers:
  - https://learn.onshape.com/courses/introduction-to-assembly-design
  - https://learn.onshape.com/courses/fundamentals-onshape-assemblies

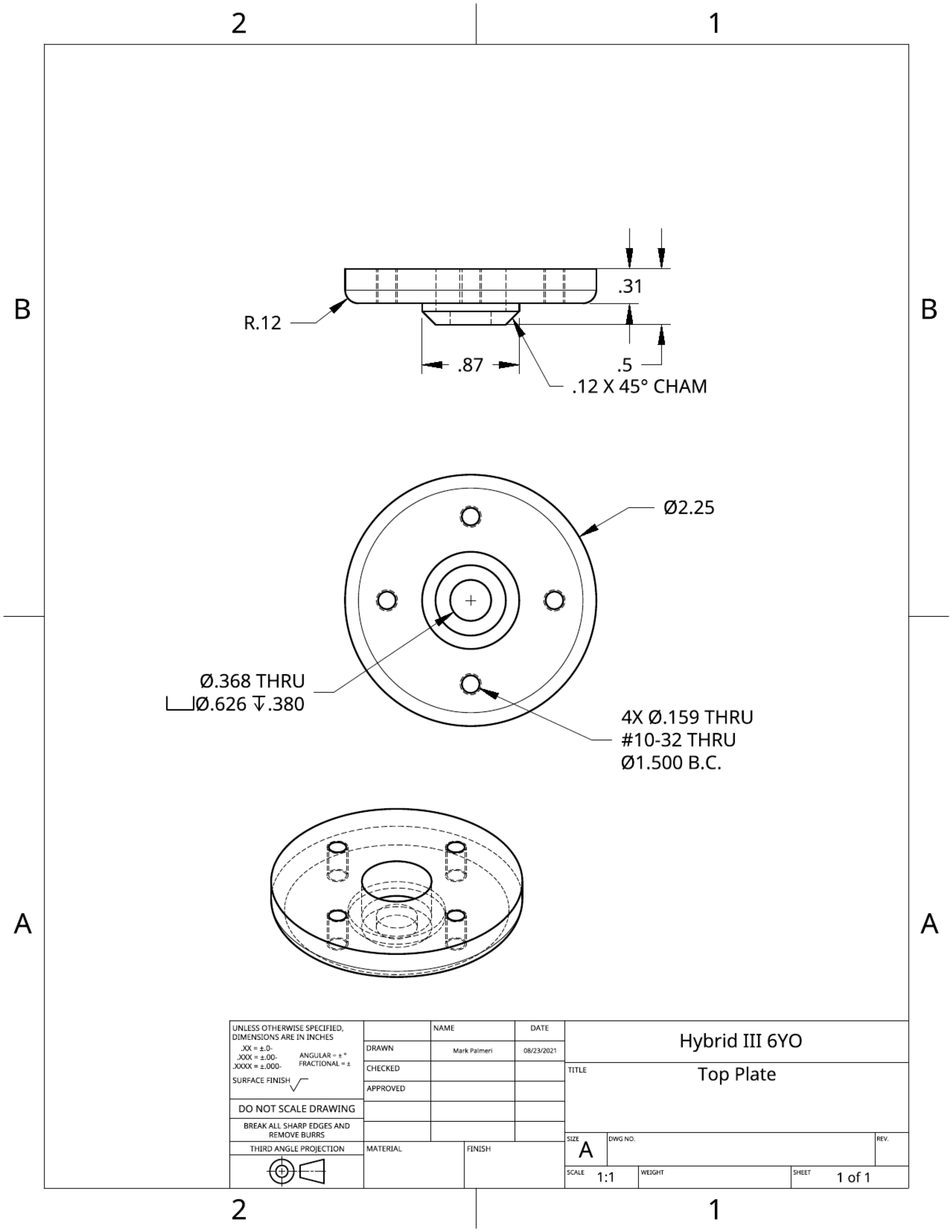
What to submit

Upload mechanical drawings and a screenshot of the part for all of your parts to the `Hybrid III 6 y/o` Gradescope assignment.

Bottom Plate

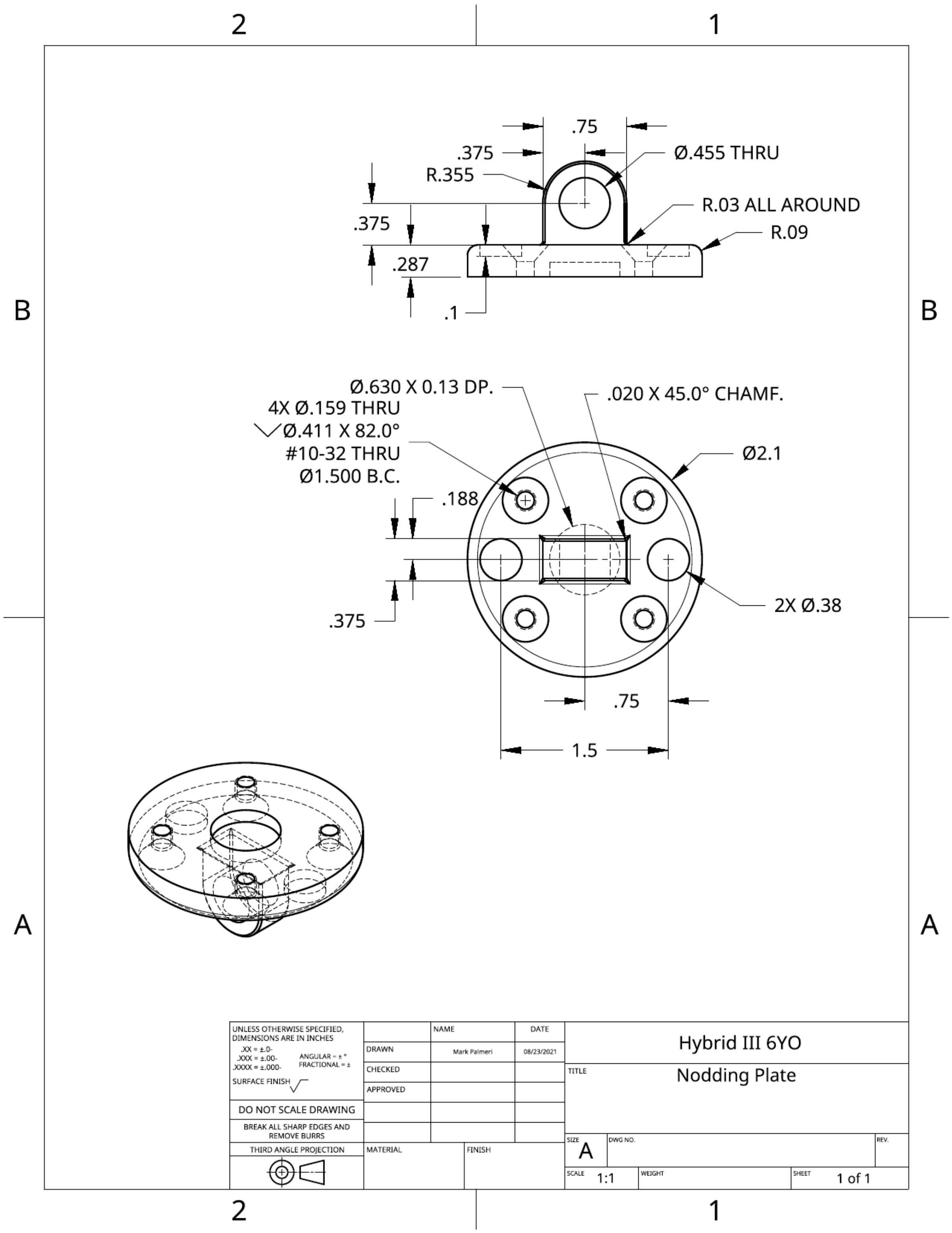


Top Plate

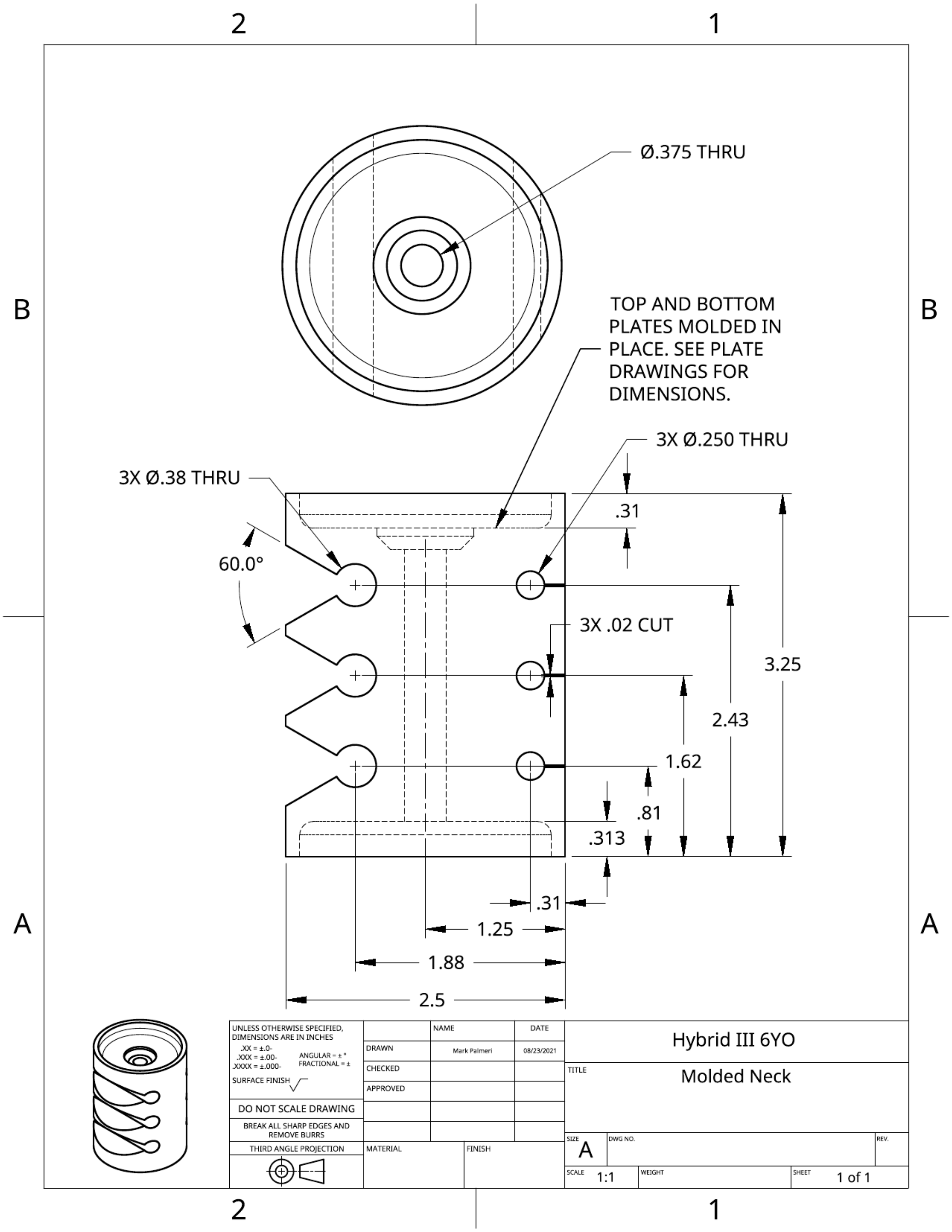


2
1
B
A
R.12
.31
.87
.5
.12 X 45° CHAM
Ø2.25
Ø.368 THRU
⌴Ø.626 ↧.380
4X Ø.159 THRU
#10-32 THRU
Ø1.500 B.C.
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
Mark Palmeri
08/23/2021
CHECKED
APPROVED
MATERIAL
FINISH
Hybrid III 6YO
TITLE
Top Plate
SIZE
A
DWG NO.
REV.
SCALE
1:1
WEIGHT
SHEET
1 of 1

Nodding Plate



Molded Neck


Ø.375 THRU
TOP AND BOTTOM PLATES MOLDED IN PLACE. SEE PLATE DRAWINGS FOR DIMENSIONS.
3X Ø.250 THRU
3X Ø.38 THRU
60.0°
.31
3X .02 CUT
3.25
2.43
1.62
.81
.313
.31
1.25
1.88
2.5
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
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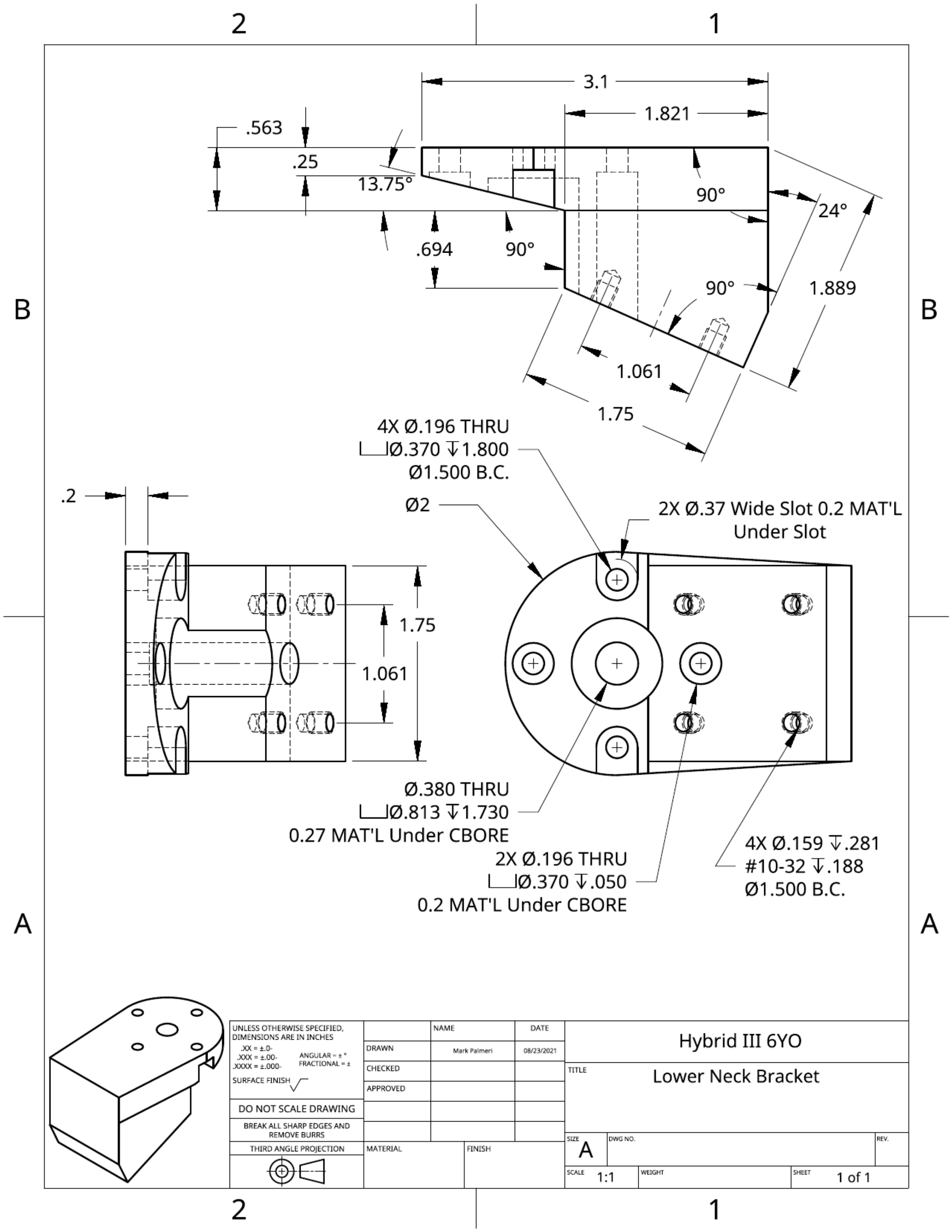
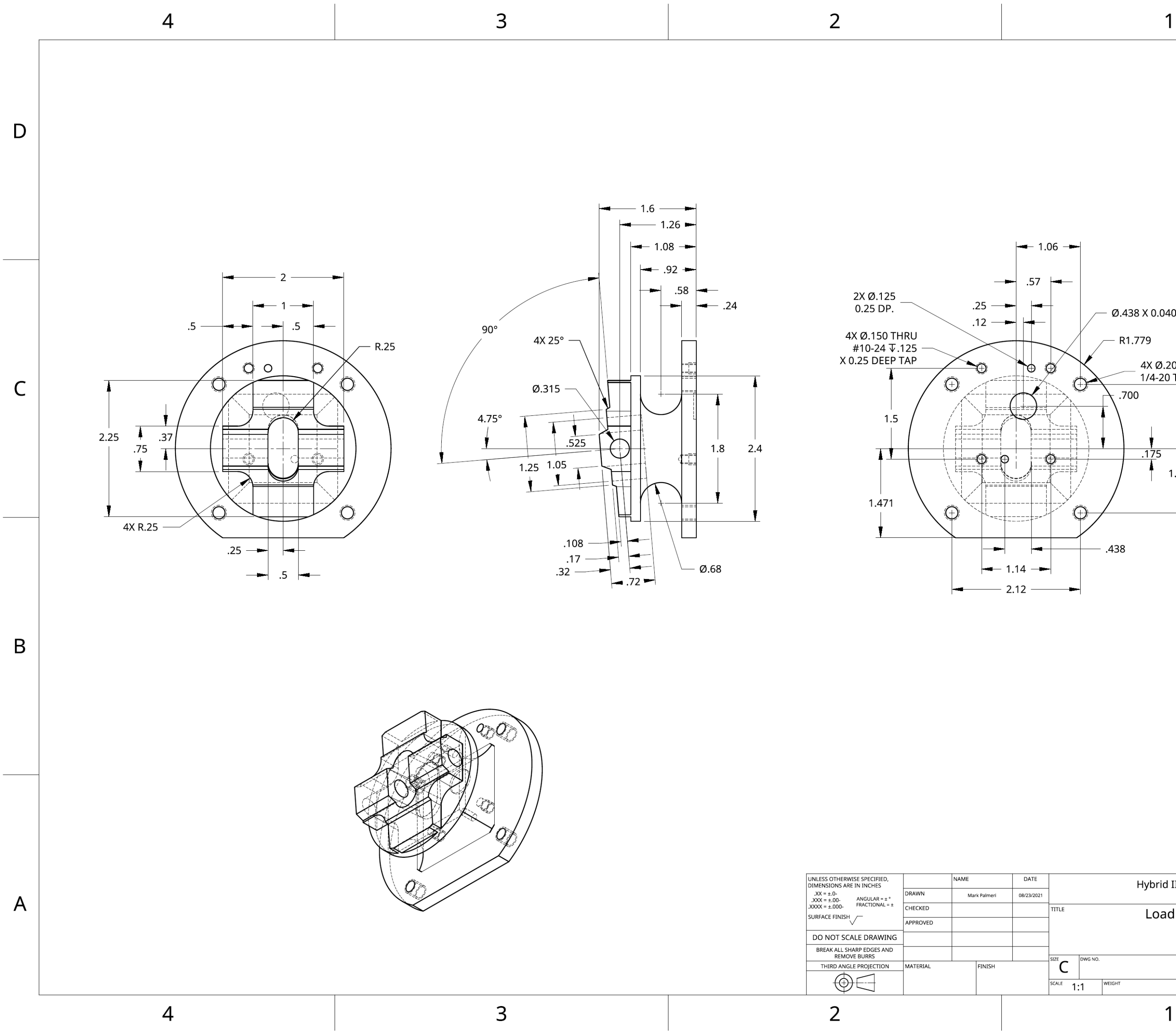
Tip

Note that the this part has features that are dependent on the top and bottom plates. To reduce the overhead of manually entering these dimensions from the other parts, you can use the `Derived` tool in Onshape to bring in the relevant faces from the other parts. https://cad.onshape.com/help/Content/derived.htm#StepsDesktop

Lower Neck Bracket



Load Cell



4
3
2
1
D
C
B
A
R.25
4X R.25
2.25
.75
.37
.25
.5
1
2
90°
4X 25°
Ø.315
4.75°
.525
1.25
1.05
1.6
1.26
1.08
.92
.58
.24
1.8
2.4
.108
.17
.32
.72
Ø.68
2X Ø.125
0.25 DP.
4X Ø.150 THRU
#10-24 ⊽.125
X 0.25 DEEP TAP
1.06
.57
.25
.12
R1.779
.700
1.5
1.471
.175
.438
1.14
2.12
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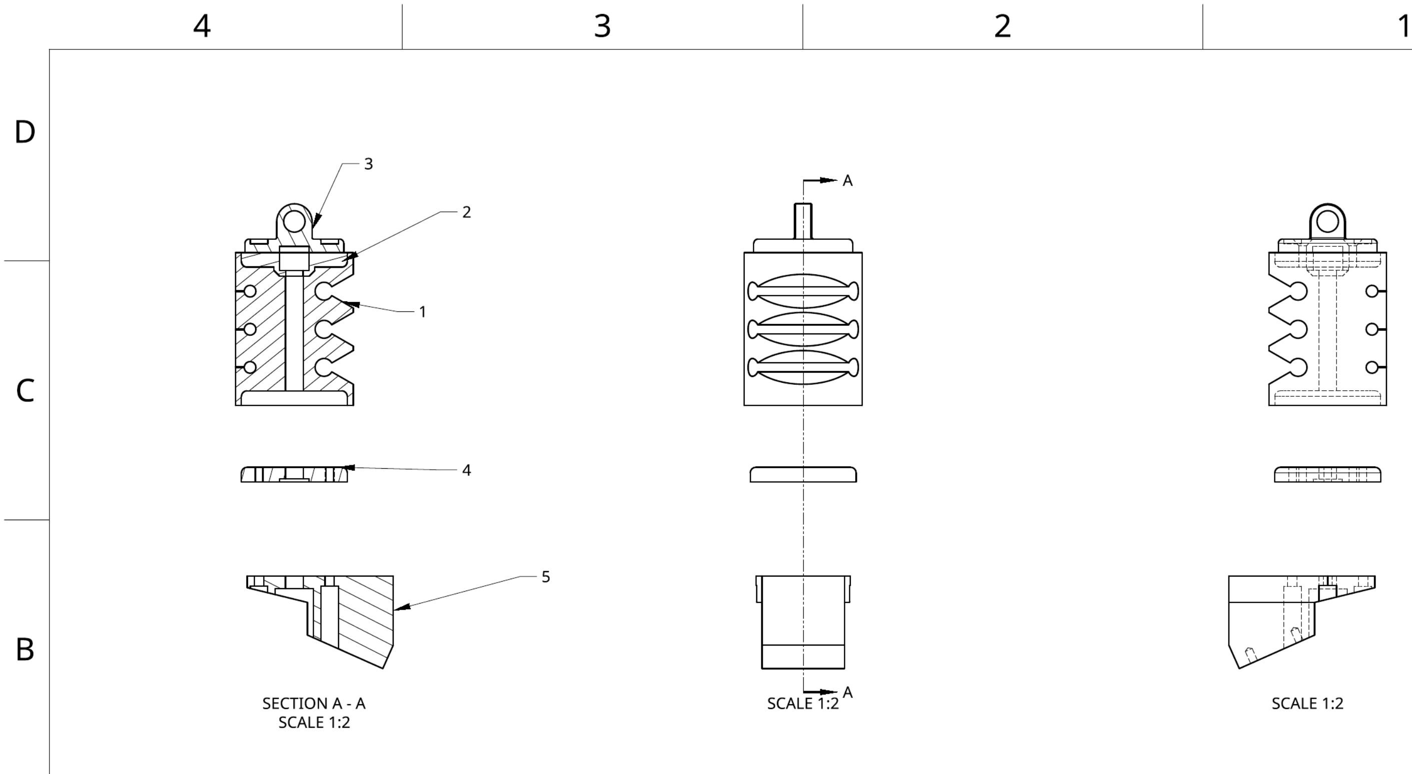
TITLE
Load
SIZE
C
DWG NO.
SCALE
1:1
WEIGHT

Assembly



Item	Quantity	Part number	Material
1	1	MOLDED NECK	Butyl
2	1	TOP PLATE	Steel
3	1	NODDING PLATE	Steel
4	1	BOTTOM PLATE	Steel
5	1	LOWER NECK BRACKET	Aluminum

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THIRD ANGLE PROJECTION

	NAME	DATE
DRAWN	Mark Palmeri	08/23/2021
CHECKED		
APPROVED		

MATERIAL
FINISH

Hybrid III 6YO

TITLE
Assembly

SIZE B
DWG NO.
SCALE 1:1
WEIGHT

4 3 2 1